Peiffer commutator and Peiffer product of precrossed B-modules in the semi-abelian context

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(joint work with A. Cigoli and S. Mantovani)

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Overview of the talk

- Intro
- Internal (pre)crossed modules
- Peiffer product
- Peiffer commutator
- A special case

Crossed Modules of groups

Let B be a group. A **precrossed B-module** $A = (a, \xi)$ is a group homomorphism a endowed with group action ξ:

$$B \times A \xrightarrow{\xi} A \xrightarrow{a} B$$

such that

$$a(\beta \cdot \alpha) = \beta + a(\alpha) - \beta$$

The pair (a, ξ) is a **crossed B-module** if it satisfies the *Peiffer Condition*

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- a useful tool in homotopical algebra, algebraic topology...
- a categorical notion:

$$\begin{array}{ccc} \mathrm{XMod} & \simeq & \mathrm{Gpd(Gp)} \\ \downarrow & & \downarrow \\ \mathrm{PXMod} & \simeq & \mathrm{RG(Gp)} \end{array}$$

(reflective subcategories)

Peiffer commutator...

Given two precrossed submodules X and Y of A:

$$\begin{tikzcd} X \arrow[r, tail] \arrow[dr] & A \arrow[d, "a"] & Y \arrow[l, tail'] \arrow[dl] \\ & B & \end{tikzcd}$$

their Peiffer commutator $\langle X, Y \rangle$ is the subgroup of A, generated by the *Peiffer words*

$$\langle x, y \rangle = x + y - x - x \cdot y \qquad \langle y, x \rangle = y + x - y - y \cdot x$$

Fact

$$\eta_A \colon \; A \twoheadrightarrow \frac{A}{\langle A, A \rangle}$$

gives the reflection $R \colon \mathrm{PXMod}_B \to \mathrm{XMod}_B$

...and Peiffer product

A related construction is the Peiffer product:

$$X \bowtie Y := \frac{X * Y}{\langle X, Y \rangle}$$

[Gilbert, Higgins '87].

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Fact

When X and Y are crossed B-modules

$$X \bowtie Y$$

is their coproduct in XMod_B

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- Problem: characterize semi-abelian categories where crossed modules condition reduces to Peiffer condition
- Answer: Martins-Ferreira, Van der Linden:

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- Problem: What is a convenient setting for working with internal crossed modules?

Some suggestions will come from the internal notion of Peiffer commutator

Internal actions

Definition

Let $\mathcal{C}$ be a semi-abelian category.

- *An internal action of an object B on an object A is an algebra $\xi\colon B\flat A \to A$ for the monad $B\flat -\colon \mathcal{C} \to \mathcal{C}$, where $B\flat A$ is the kernel of*

$$[1,0]\colon B + A \to B$$

- *The (canonical) conjugation action of an object B on itself is given by the composition:*

$$B\flat B \longrightarrow B + B \xrightarrow{[1,1]} B,$$

(Pre-)crossed modules in a (SH) semi-abelian category

Let $\mathcal{C}$ a semi-abelian category, and B an object of $\mathcal{C}$.

(Pre-)crossed modules in a (SH) semi-abelian category

Let $\mathcal{C}$ a semi-abelian category, and B an object of $\mathcal{C}$.

An precrossed B-module in $\mathcal{C}$ is a pair $B\flat X \xrightarrow{\xi} X \xrightarrow{x} B$ *, (ξ an internal action), such that:*

$$\text{s.t.} \quad \begin{tikzcd} B\flat X \arrow[r, "\xi"] \arrow[d, "1\flat x"'] \arrow[dr, phantom, "\equiv"] & X \arrow[d, "x"] \\ B\flat B \arrow[r, "\chi"'] & B \end{tikzcd}$$

If $\mathcal{C}$ satisfies the property (SH), (x, ξ) is crossed B-module iff

$$\begin{tikzcd} X\flat X \arrow[r, "\chi"] \arrow[d, "x\flat 1"'] \arrow[dr, phantom, "\equiv"] & X \arrow[d, equal] \\ B\flat X \arrow[r, "\xi"'] & X \end{tikzcd}$$

The category $\mathrm{PXMod}_B(\mathcal{C})$

$$\mathrm{PXMod}_B(\mathcal{C}) = \left\{ \begin{tikzcd} X \arrow[d, "x"] \\ B \end{tikzcd} , \begin{tikzcd} Y \arrow[d, "y"] \\ B \end{tikzcd} , \cdots \quad \begin{tikzcd} X \arrow[rr, "f"] \arrow[dr, "x"'] & & Y \arrow[dl, "y"] \\ & B & \end{tikzcd} , \cdots \right\}$$

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Fact

$\mathrm{PXMod}_B(\mathcal{C})$ *is*

- *finitely complete and finitely cocomplete*
- *quasi pointed:* $0 \rightarrowtail 1$
- *Barr-exact*
- *protomodular*

The normalization functor

$$\mathsf{RG}_B(\mathcal{C}) \to \mathsf{PXMod}_B(\mathcal{C})$$

behaves nicely w.r.t.

- taking limits
- regular epi/mono factorization

Fact

Finite limits and the reg. epi/mono factorization are computed at the level of domains.

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Fact

Finite limits and the reg. epi/mono factorization are computed at the level of domains.

Colimits are more involved.

Peiffer product

Consider $(x\colon X \to B, \xi_X)$ and $(y\colon Y \to B, \xi_Y)$ in $\mathrm{PXMod}_B(\mathcal{C})$.

Proposition

The (unique) arrow $[x, y\rangle\colon X \rtimes_{y*\xi_X} Y \to B$ *making the diagram commute*

$$\begin{tikzcd}
X \arrow[dr, "j_X"] \arrow[ddr, "x"'] & & Y \arrow[dl, "i_Y"'] \arrow[ddl, "y"] \\
& X \rtimes_{y*\xi_X} Y \arrow[d, "{[x,y\rangle}"] & \\
& B &
\end{tikzcd}$$

is a precrossed module, and j_X *and* i_Y *induce morphisms in* $\mathrm{PXMod}_B(\mathcal{C})$.

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$$\begin{tikzcd}
X \arrow[r] \arrow[dr, "j_X"] \arrow[ddr, "x"'] & X +_{\mathrm{PX}} Y \arrow[d, "{[j_X, i_Y]_{\mathrm{PX}}}"] & Y \arrow[l] \arrow[dl, "i_Y"'] \arrow[ddl, "y"] \\
& X \rtimes_{y*\xi_X} Y \arrow[d, "{[x,y\rangle}"] & \\
& B &
\end{tikzcd}$$

is a precrossed module, and j_X *and* i_Y *induce morphisms in* $\mathrm{PXMod}_B(\mathcal{C})$.

By UP of coproducts in $\mathrm{PXMod}_B(\mathcal{C})$, we get the morphism $[j_X, i_Y]_{\mathrm{PX}}$.

Let us consider this morphism $[x, y\rangle$:

$$\begin{tikzcd}
X +_{\mathrm{PX}} Y \arrow[r, "{[j_X, i_Y]_{\mathrm{PX}}}"] \arrow[d, "{[x,y]}"'] & X \rtimes Y \arrow[dl, "{[x,y\rangle}"] \\
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\end{tikzcd}$$

Let us consider this morphism $[x, y\rangle$:

$$\begin{tikzcd}
X \,{}_{\xi}\!\square\, Y \arrow[r, tail] \arrow[dr, "0"'] & X +_{\mathrm{PX}} Y \arrow[r, "{[j_X, i_Y]_{\mathrm{PX}}}"] \arrow[d, "{[x,y]}"'] & X \rtimes Y \arrow[dl, "{[x,y\rangle}"] \\
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Its kernel is denoted by $X \,{}_{\xi}\!\square\, Y$

Symmetrically, one defines $X \,\square_{\xi}\, Y = \mathrm{Ker}([j_Y, i_X]_{\mathrm{PX}})$.

Definition

$$X \underset{\xi}{\square} Y := (X \,{}_{\xi}\!\square\, Y) \vee_{\mathrm{PX}} (X \,\square_{\xi}\, Y)$$

Since $\mathrm{PXMod}_B(\mathcal{C})$ is sequentiable, $X \underset{\xi}{\square} Y$ is the kernel of the (diagonal of the) following pushout in $\mathrm{PXMod}_B(\mathcal{C})$:

$$\begin{tikzcd}
X +_{\mathrm{PX}} Y \arrow[r, "{[j_X, i_Y]_{\mathrm{PX}}}"] \arrow[d, "{[i_X, j_Y]_{\mathrm{PX}}}"'] & X \rtimes Y \arrow[d] \\
Y \rtimes X \arrow[r] & X \bowtie Y
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Definition

$X \bowtie Y$ is the **internal Peiffer product** *of x and y.*

The Peiffer product comes with two canonical morphisms in $\mathrm{PXMod}_B(\mathcal{C})$:

$$X \xrightarrow{\overline{x}} X \bowtie Y \xleftarrow{\overline{y}} Y$$

These are obtained by precomposition with coproduct injections.

Peiffer commutator

Consider X and Y subobjects of A in $\mathrm{PXMod}_B(\mathcal{C})$:

$$\begin{tikzcd} X \arrow[r, tail, "f"] \arrow[dr, "x"'] & A \arrow[d, "a"] & Y \arrow[l, tail, "g"'] \arrow[dl, "y"] \\ & B & \end{tikzcd}$$

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Definition

The **Peiffer commutator** $\langle X, Y \rangle$ *is defined by the factorization:*

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Proposition

Given a cospan of subobjects in $\mathrm{PXMod}_B(\mathcal{C})$*, T.F.A.E.*

1. $\langle X, Y \rangle = 0$;
2. $$\begin{tikzcd} X \arrow[dr, tail, "f"'] & & Y \arrow[dl, tail, "g"] \\ & A & \end{tikzcd}$$

Proposition

Given a cospan of subobjects in $\mathrm{PXMod}_B(\mathcal{C})$*, T.F.A.E.*

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2. *there exists a (unique) morphism* φ *such that the diagram commutes.*

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X \arrow[r, "\overline{x}"] \arrow[dr, tail, "f"'] & X \bowtie Y \arrow[d, dashed, tail, "\varphi"] & Y \arrow[l, "\overline{y}"'] \arrow[dl, tail, "g"] \\
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Consequence:

$$A \quad \text{crossed module} \quad \Rightarrow \quad \langle A, A \rangle = 0$$

For any pair of precrossed sub-modules X and Y

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Let $\mathcal{C}$ be semi-abelian such that condition (SH) holds.

Theorem

The following quotient gives the reflection $\mathrm{PXMod}_B(\mathcal{C}) \to \mathrm{XMod}_B(\mathcal{C})$*:*

$$\begin{tikzcd}
\langle X, X\rangle \arrow[r, tail] \arrow[dr, "0"'] & X \arrow[r, "\eta_X"] \arrow[d, "x"] & \dfrac{X}{\langle X, X\rangle} \arrow[dl, "R(x)"] \\
& B &
\end{tikzcd}$$

Idea of the proof:

- Peiffer condition

$$\left\langle \frac{X}{\langle X, X\rangle}, \frac{X}{\langle X, X\rangle} \right\rangle = 0$$

- UP of the unit

$$\begin{tikzcd}
X +_{PX} X \arrow[r, "{[1,1]_{PX}}"'] \arrow[d] & X \arrow[d, "\eta_X"'] \arrow[ddr, bend left, "\forall f"] & \\
X \bowtie X \arrow[r] \arrow[drr, bend right, "\exists \text{ since } \langle f(X), f(X)\rangle = 0"'] & \dfrac{X}{\langle X, X\rangle} \arrow[dr] & \\
& & A \in \mathrm{XMod}_B(\mathcal{C})
\end{tikzcd}$$

The Algebraically Coherent case

Let $\mathcal{C}$ be a semi-abelian category satisfying the condition of Algebraic Coherence, in the sense of Cigoli, Gray, Van der Linden:

(AC) *Kernel functors preserve jointly strongly epic pairs*

(examples: groups, rings, Lie algebras, Leibniz algebras, any category of interest in the sense of Orzech...)

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Immediately one gets

Proposition

The canonical map

$$X + Y \to X +_{\mathrm{PX}} Y$$

is a regular epimorphism

In this case

$$\begin{tikzcd}
X + Y \arrow[r, two heads] \arrow[d, two heads, "{[j_X, i_Y]}"'] & X +_{\mathrm{PX}} Y \arrow[d, two heads, "{[j_X, i_Y]_{\mathrm{PX}}}"] \\
X \rtimes Y \arrow[r, equal] & X \rtimes Y
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In this case

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 & X \,{}_{\xi}\!\square\, Y \arrow[d, tail] \\
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In this case

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X \,{}_{\xi}\!\Diamond\, Y \arrow[r, two heads] \arrow[d, tail] & X \,{}_{\xi}\!\Box\, Y \arrow[d, tail] \\
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Symmetrically: $X \Diamond_{\xi} Y \twoheadrightarrow X \Box_{\xi} Y$

$$X \underset{\xi}{\Diamond} Y := (X \,{}_{\xi}\!\Diamond\, Y) \vee (X \Diamond_{\xi} Y)$$

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Fact

$$X \underset{\xi}{\Diamond} Y \twoheadrightarrow X \underset{\xi}{\Box} Y$$

is a regular epimorphism

- the Peiffer product can be obtained as a pushout in $\mathcal{C}$:

$$\begin{tikzcd} X +_{\mathrm{PX}} Y \arrow[r] \arrow[d] & X \rtimes Y \arrow[d] \\ Y \rtimes X \arrow[r] & X \bowtie Y \end{tikzcd}$$

- the Peiffer product can be obtained as a pushout in $\mathcal{C}$:

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- the Peiffer product can be obtained as a pushout in $\mathcal{C}$:

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& Y \rtimes X \arrow[r] & X \bowtie Y
\end{tikzcd}$$

- for any pair of subobjects in $\mathrm{PXMod}_B(\mathcal{C})$, $\langle X, Y \rangle$ is computed:

$$\begin{tikzcd}
X \underset{\xi}{\square} Y \arrow[r, two heads] \arrow[d, tail] & \langle X, Y \rangle \arrow[d, tail] \\
X +_{\mathrm{PX}} Y \arrow[r, "{[f,g]_{\mathrm{PX}}}"] & A
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X + Y \arrow[r, two heads] \arrow[rr, bend right, two heads, "{[f,g]}"'] & X +_{\mathrm{PX}} Y \arrow[r, two heads, "{[f,g]_{\mathrm{PX}}}"] & A
\end{tikzcd}$$

- Advantage: need not to compute colimits in $\mathrm{PXMod}_B(\mathcal{C})$.

When we consider $X = (0, \tau_X)$ and $Y = (0, \tau_Y)$ precrossed modules endowed with trivial actions, we observe:

$$X \times Y = X \rtimes Y = X \ltimes Y$$

- the definition of $X \bowtie Y$ becomes

$$\begin{tikzcd} X + Y \arrow[r, two heads] \arrow[d, two heads] & X \times Y \arrow[d, equal] \\ X \times Y \arrow[r, equal] & X \times Y \end{tikzcd}$$

- ...so that $X \underset{\xi}{\Diamond} Y = X \diamond Y$ and the Peiffer commutator coincides with the Higgins commutator [Mantovani, M. - 2010]:

$$\begin{tikzcd} X \diamond Y \arrow[r, two heads] \arrow[d, tail] & {[X, Y]_H} \arrow[d, tail] \\ X + Y \arrow[r, "{[f,g]}"'] & A \end{tikzcd}$$

Universal Property of $\bowtie$

In the (AC) case we have a UP of $X \bowtie Y$

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\begin{tikzcd}
 & Y \flat X \arrow[dl, "y^*\xi_X"'] \arrow[dr] & & X \flat Y \arrow[dl] \arrow[dr, "x^*\xi_Y"] & \\
X \arrow[dr] & & X + Y \arrow[dl] \arrow[dr] & & Y \arrow[dl] \\
 & X \rtimes Y \arrow[dr] & & Y \ltimes X \arrow[dl] & \\
 & & X \bowtie Y & &
\end{tikzcd}
$$

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X \arrow[drr, dashed] & & X + Y \arrow[d, dashed] & & Y \arrow[dll, dashed] \\
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X \arrow[drr, dashed] \arrow[ddrr] & & X + Y \arrow[d, dashed] & & Y \arrow[dll, dashed] \arrow[ddll] \\
 & & X \bowtie Y \arrow[d] & & \\
 & & A & &
\end{tikzcd}
$$

"X et Y Peiffer-commute iff their reciprocally induced actions turn into coniugation inside A"

Thank you for your attention!